Name: Evangelista, VJ P.
Section: BSIT 2-1

NO.:
DATE:

Evangelista, VJ P. COMP 009 - Object Oriented Programming
BSIT 2-1 Midterm Activity #1 - Variable

PART A - Variable Identification Table

Variable Name	Data Type	Variable Type	Stored In
name	String (non-primitive)	instance	heap
grade	int (primitive)	instance	heap
n	String (non-primitive)	local (parameter)	stack
g	int (primitive)	local (parameter)	stack
s1	Student (constructor)	local	stack (reference), heap (object)
s2	Student (constructor)	local	stack (reference), heap (object)
s3	Student (constructor)	local	stack (reference), heap (object)

PART B - Explanation. Please see example below.

Statement	What Happens?	Memory Impact
Student s1 = new Student ("John", 85);	The value "John", 85 will be assigned in s1 using the constructor Student.	s1 is stored on a stack and adding object on a heap.
Student s2 = new Student ("Maria", 90);	the value "Maria", 90 will be assigned in s2 using the constructor Student.	s2 is stored on a stack and adding object on a heap
Student s3 = s2;	the value "Maria, 90" will also be assigned to s3, as s3 have the same value as s2 using the constructor Student.	s3 is stored on a stack but referred the same object on s2
s1.grade = 95;	It changes the value of s1.grade from 85 to 95.	the value that replaced s1.grade is stored on heap.
s3.name = "Ana";	s2.name will be changed to "Ana" as s3 have the same value as s2, thus s2 will also be changed.	the value that replaced s2.name is stored on stack.

VICTORY

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PART C - OUTPUT

John, 95

Ana, 90

Ana, 90